W-L-124a
October 24, 1966

SUPERSEDING
Int. Fed. Spec. W-L-00124(GSA-FSS)
April 14, 1965

FEDERAL SPECIFICATION

LIGHT, VEHICULAR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers vehicular lights accommodating incandescent filament sources within an enclosure for operation on commercial vehicles. Includes only stop, directional, tail, clearance marker lights, and combinations thereof.

1.2 Classification. The vehicular lights furnished under this specification shall be of the following types and classes, as specified (see 6.2).

Type:
- I—Stop-directional.
- II—Tail and stop-directional.
- III—Marker, clearance.
- IV—Tail.

Class:
- A—Amber (yellow) lens.
- B—Red lens.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-349—Plastic Compounds, Molding, Cellulose Acetate Butyrate; and Molded or Extruded Parts.
L-P-380—Plastic Molding Material, Methacrylate.
L-P-392—Plastic Molding Material, Acetal, Injection and Extrusion.
L-P-393—Plastic Molding Material, Polycarbonate, Injection and Extrusion.
W-L-111—Lamp, Incandescent, (Electric, Miniature, Tungsten-Filament).
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-640—Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320—Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and

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Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

The Society of Automotive Engineers (SAE), Inc. Standards:

J 585—Tail Lamps.
J 586—Stop Lamps.
J 588—Turn Signal Lamps.
J 592—Clearance, Side Marker, Identification, and Parking Lamps.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 485 Lexington Ave., New York, N. Y., 10017.)

American Society for Testing and Materials (ASTM) Publication:

ASTM D 1788—Rigid Acrylonitrile-Butadiene-Styrene (ABS) Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 General.

3.1.1 The vehicular lights shall be so designed and constructed that they will satisfactorily meet the lighting service conditions encountered after installation on the vehicle. They shall be equipped with a lens, or lenses, made of an optic material capable of diffusing or otherwise redirecting the light flux from the enclosed light source to meet this requirement. The incandescent lamp, when furnished with the vehicular light, shall comply with the requirements of W-L-111. Unless otherwise specified (see 6.2), the lights shall be suitable for use on 12-volt electrical systems.

3.1.2 *Plastic material.* All plastic material used for the items covered by this specification shall conform to the applicable requirements of L-P-349, L-P-380, L-P-392, L-P-393, or ASTM D 1788.

3.2 Design.

3.2.1 The vehicular lights shall be dust, moisture, corrosion, and vibration resistant.

3.2.2 The incandescent lamp mounting used in the vehicular lights shall be prepositioned to insure the lamp filament orientation to the optical element is such as to constantly meet the photometric requirements specified.

3.2.3 The terminals shall be copper or copper alloy with a corrosive resistant finish to assure good electrical contact.

3.3 Lens.

3.3.1 *Material.* Unless otherwise specified (see 6.2), the lens shall be of the uncoated glass or plastic type at the option of the manufacturer.

3.3.2 *Color.* The color of the lens shall conform to the applicable SAE standard. The colors shall be as specified in 1.2 of this specification.

3.3.3 *Warpage of plastic lens.* Where used, there shall be no warpage of plastic

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lenses which would affect the proper function of the vehicular light.

3.4 Dimensions and configuration. Unless otherwise specified (see 6.2), the dimensions and configuration of the vehicular lights furnished under this specification shall be in accordance with SAE standards where applicable.

3.5 Performance.

3.5.1 *Dust.* When subjected to the dust test (see 4.3.1), the vehicular light shall show no evidence of a loss of more than 10 percent in maximum candlepower.

3.5.2 *Vibration.* When subjected to the vibration test (see 4.3.1), lights shall show no evidence of material physical weakness, lens or reflector rotation, displacement or rupture of parts.

3.5.3 *Moisture.* When subjected to the moisture test (see 4.3.1), the vehicular light shall show no evidence of more than 2 cc. accumulation within the light.

3.5.4 *Corrosion.* When subjected to the corrosion test (see 4.3.1), the vehicular light shall show no evidence of excessive corrosion that would affect the proper functioning of the device.

3.5.5 *Photometric qualities.*

3.5.5.1 *Type I.* The type I vehicular light shall have photometric qualities as set forth in SAE J-586 and J-588, as applicable.

3.5.5.2 *Type II.* The type II vehicular light shall have photometric qualities as set forth in SAE J-585, J-586, and J-588, as applicable.

3.5.5.3 *Type III.* The type III vehicular light shall have photometric qualities as set forth in SAE J-592, as applicable.

3.5.5.4 *Type IV.* The type IV vehicular light shall have photometric qualities as set forth in SAE J-585, as applicable.

3.6 Workmanship. The vehicular lights shall be free from characteristics or defects which affect the appearance, or which might affect the serviceability or render the vehicular light unsuitable or inefficient for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Sampling for visual inspection shall be conducted in accordance with MIL-STD 105 at inspection level S-3. Acceptance shall be based upon an acceptable quality level (AQL) of 6.5.

4.3 Tests.

4.3.1 The color, warpage, dust, vibration, moisture, corrosion, and photometric test shall be made in accordance with the applicable requirements of SAE J-585, J-586, J-588 and J-592, respectively. (See 3.3.2, 3.3.3, and 3.5.1 through 3.5.5).

4.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to

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the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-3 with an AQL of 3.5 defects per hundred units.

TABLE I. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling.
	Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Six lights of one type and class shall be packaged in a box conforming to PPP-B-636, class weather-resistant. Interior corrugated fiberboard partitions with the fiberboard meeting the requirements of PPP-F-320, class domestic, variety SW shall be used in each box to form individual cells for each lamp unit. Closure shall be in accordance with the appendix to the box specification.

5.1.2 *Level B.* Six lights of one type and class shall be packaged in a box conforming to PPP-B-636, type CF, class domestic. Interior corrugated fiberboard partitions with the fiberboard meeting the requirements of PPP-F-320, class domestic, variety SW shall be used in each box to form individual cells for each lamp unit. Closure shall be in accordance with the appendix to the box specification.

5.1.3 *Level C.* Lights shall be packaged in a manner which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the receiving activity at destination. The supplier's commercial practice will be acceptable when it fulfills the above requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Lights of one type and class, shall be packed in a close-fitting box conforming to PPP-B-601 overseas type; PPP-B-621, class 2; PPP-B-636, class weather-resistant or PPP-B-640, class 2, grade A. The box shall be closed and strapped in accordance with the appendix to the applicable box specification. Each fiberboard box shall be waterproof sealed with tape in accordance with the appendix to PPP-B-636 when packed in the wood boxes.

5.2.2 *Level B.* Twelve unit packages of lights of one type and class shall be packed in a box conforming to PPP-B-636, type CF, class domestic. Closure shall be in accordance with the appendix to the box specification.

5.2.3 *Level C.* The lights shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special markings required by the contract or order the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special markings required by the contract or order the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The vehicular lights furnished under this specification are intended

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for use on various types of trucks and trailers. They are not intended for use on passenger vehicles.

6.2 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Type and class required (1.2).

(*c*) Electrical system, if other than specified (3.1.1).

(*d*) Material, if other than specified (3.3.1).

(*e*) Dimensions and configuration, if other than specified (3.4).

(*f*) Inspection responsibility, if other than specified (4.1).

(*g*) Selection of applicable levels of packaging and packing required (5.1 and 5.2).

MILITARY INTERESTS:

CUSTODIANS:

Army—MO

Navy—YD

Air Force—84

Reviewers:

Army—MO

Navy—YD

Air Force—84, 85

Users:

Army—WC, SM, CE

Navy—WP, MC

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2, of this specification to obtain extra copies and other documents referenced herein. Price [illegible] cents each.

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